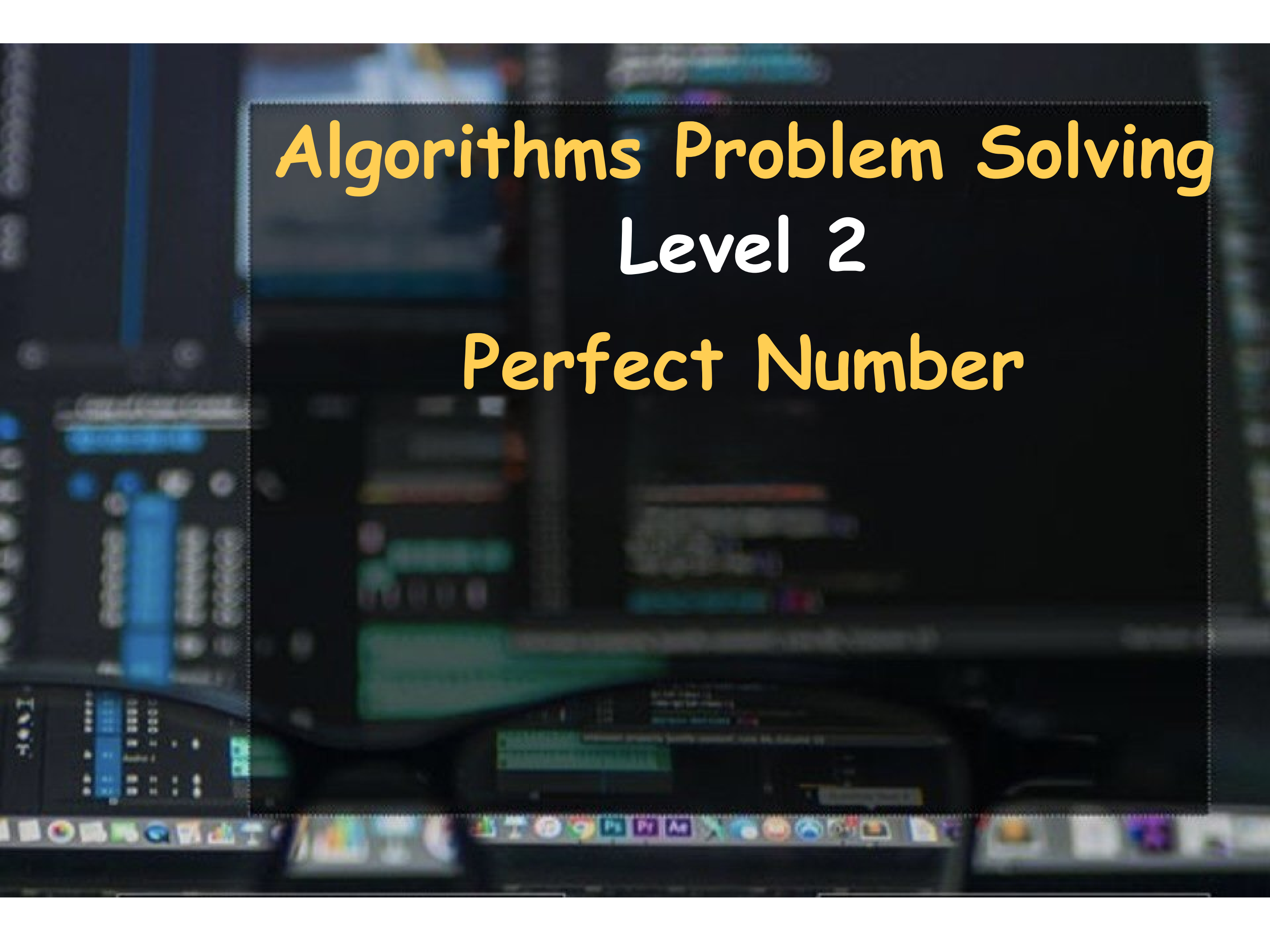
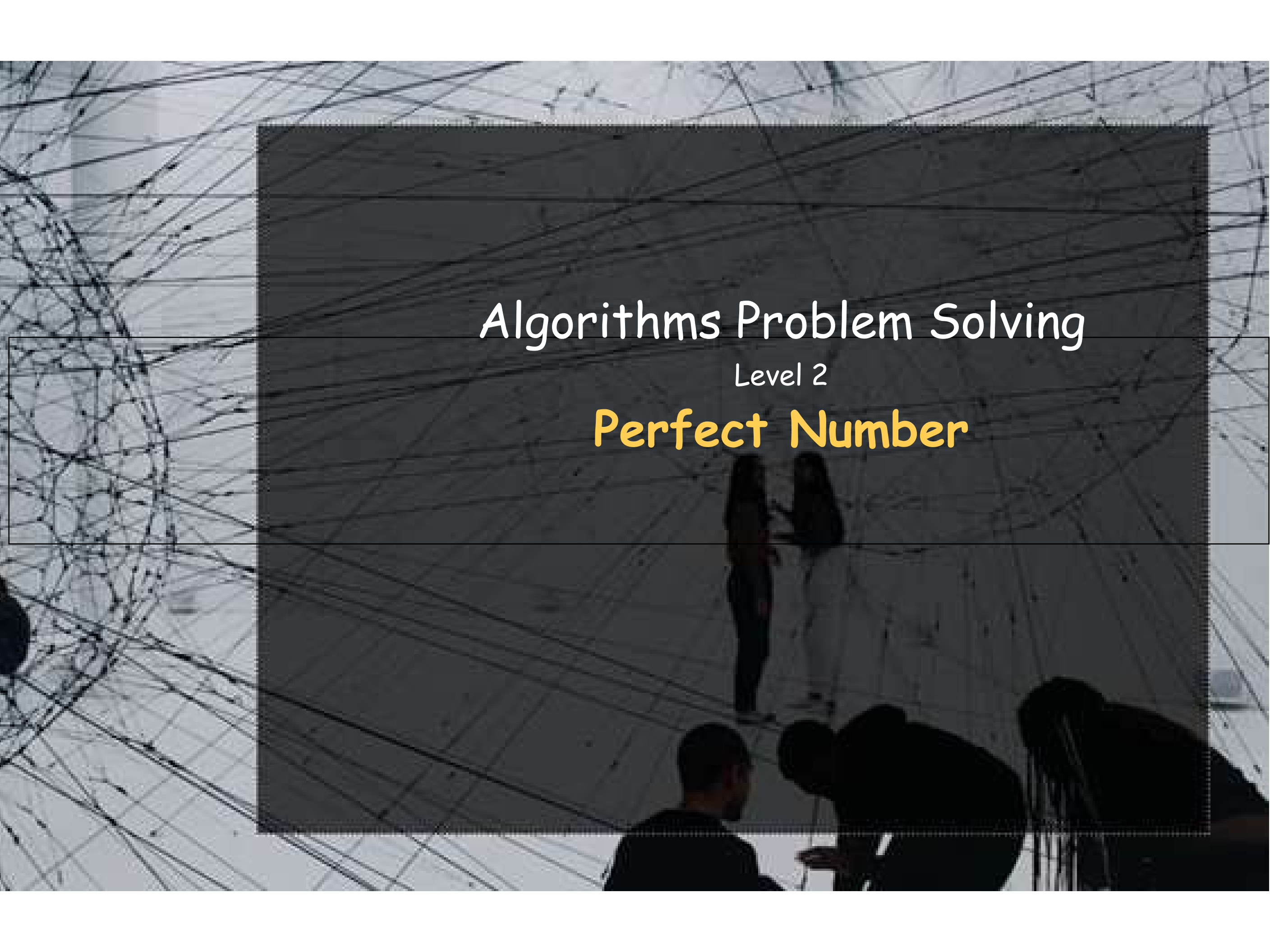




# Algorithms Problem Solving

## Level 2

### Perfect Number



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## Perfect Number

## Definition

◆ A Perfect Number is a positive number that is equal to the sum of all its proper divisors, excluding the number itself.

## Explanation

- ◆ ``ReadNumber()`` reads a positive number from the user using a ``do-while`` loop.
- ◆ ``CheckPerfect()`` calculates the sum of all divisors of the number except the number itself.
- ◆ The ``for`` loop starts from ``1`` and continues while ``i < Numb``.
- ◆ ``Numb % i == 0`` checks whether ``i`` is a divisor of ``Numb``.
- ◆ If it is a divisor, it is added to ``Sum``.
- ◆ Finally, ``Numb == Sum`` determines whether the number is perfect.
- ◆ ``PrintResult()`` prints whether the number is Perfect or Not Perfect.

## Steps in main()

1. Read a positive number.
2. Start a loop from 1 to `Numb - 1`.
3. Check if each number is a divisor using `%`.
4. Add every divisor to `Sum`.
5. Compare `Sum` with the original number.
6. If they are equal, the number is Perfect.

# 💡 Example

◆ Input:

`28`

◆ Divisors of 28:

`1 + 2 + 4 + 7 + 14 = 28`

➡ Result

`28 Is Perfect.`

## Note

- ◆ Only proper divisors are included in the sum, so the number itself is not included.
- ◆ `6` is another Perfect Number because  $1 + 2 + 3 = 6$ .

## ⊘ Warning

- ◆ The program accepts positive numbers, including `1`, but `1` is not a Perfect Number because its proper divisor sum is `0`.
- ◆ The loop checks every number from `1` to `Numb - 1`, which is simple but can be inefficient for very large numbers.



# Comparison

## ▨ Perfect Number

- ✓ Sum of all proper divisors equals the number.
- ✓ Example:  $6 \rightarrow 1 + 2 + 3 = 6$
- ✓ Example:  $28 \rightarrow 1 + 2 + 4 + 7 + 14 = 28$

## ▤ Not Perfect Number

- ✓ Sum of proper divisors does not equal the number.
- ✓ Example:  $10 \rightarrow 1 + 2 + 5 = 8$

## ✦ Summary

- ✓ A Perfect Number equals the sum of its proper divisors.
- ✓ `%` is used to find divisors.
- ✓ The number itself is excluded from the sum.
- ✓ `CheckPerfect()` returns `true` when the number is Perfect.
- ✓ `6` and `28` are examples of Perfect Numbers.

Thank you 😊